



Knowledge boundary spanning and productivity in information systems support community



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ARTICLE INFO

Article history:

Received 4 January 2014

Received in revised form 9 August 2015

Accepted 22 September 2015

Available online 30 September 2015

Keywords:

Information systems support

Knowledge boundary

Experiential learning

Boundary spanning

Productivity

Knowledge worker

ABSTRACT

An information systems (IS) support community represents a knowledge-intensive network where IS professionals interact with end-users to resolve system use problems during the IS post-implementation stage. For IS professionals in the support function, providing support for multiple information systems to multiple business units requires knowledge about different domains (business units or technical systems). In this paper, we take a network perspective to empirically evaluate the effects of an IS worker's network position and knowledge boundary spanning on productivity. Drawing upon theories of experiential learning and knowledge boundary, we perform social network analysis and linear mixed effects modeling to analyze archival data comprising 36 IS workers and 23,450 support requests made by 4568 end-users during the first 13 months post SAP/R3 implementation in a large U.S. organization. Our findings reveal that IS workers' network centrality and boundary spanning positively influence productivity. Surprisingly, their boundary-spanning experience plays a substantially more important role than network centrality. This study makes important contributions to theory and practice in individual experiential learning in knowledge-intensive networks.

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1. Introduction

Organizational support for a newly-installed information system (IS) is a knowledge-intensive setting where IS professionals engage in frequent knowledge sharing activities with their business colleagues and provide them with information, knowledge and solutions in a timely manner. Such an IS support community comprises of both end-users (business employees) who use the system to accomplish their work and IS professionals (support personnel) who provide support for system use and maintenance. We consider IS professionals in such a context knowledge workers, as they utilize their knowledge and expertise to support organizational use of the implemented systems. As such, their ability to share knowledge effectively with business employees in an organization is critical to the efficiency of IS support operation. For example, prompt and effective resolutions by IS support personnel enable employees to perform their IS-enabled work, promoting organizational use of IS [7] and facilitating the extended use of IS [24]. In contrast, their failure to efficiently address users' requests is found to cause user frustration and disruption in work [44]. Moreover, the knowledge required to efficiently perform IS support work cannot simply be a body of knowledge to be learned once. Rather, it is acquired through a dynamic process of learning from work-based activities dedicated to the support tasks at hand. Thus, understanding the factors

contributing to IS workers' learning and productivity in the support function becomes an important strategic issue for IS managers and business executives.

One important predictor for a knowledge worker's productivity is experiential learning, learning from individual experience. A worker's cumulative experience provides the worker with an opportunity to become more proficient in individual tasks and in performing established routines and practices [2,42]. As knowledge workers increasingly perform tasks in an interconnected environment, their experiential learning is likely to be affected by the network in which they are embedded. In a network, individuals not only learn through their personal experiences, but also benefit from knowledge accumulated by others [23,42]. In the IS support environment, collective learning and problem-solving among IS workers and users underlies a crucial knowledge sharing process [36,46]. In this regard, IS workers' network positions in the IS support community are likely to influence their access to knowledge sources, subsequently affecting their performance in completing IS support tasks. This suggests that examining experiential learning from a network perspective may provide insights into IS workers' productivity.

Yet, experiential learning in a knowledge network may be constrained by the boundaries of knowledge domains, such as business areas or technical systems. Knowledge boundaries often arise around specialized domains of knowledge and give rise to semantic, syntactic and cognitive differences among individuals from different knowledge domains [8,10]. Moreover, knowledge boundaries cause discontinuities

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